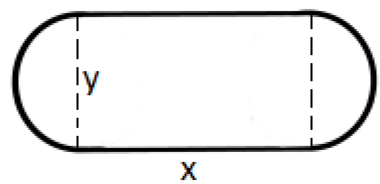
MTH 200 **Optimization**

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. A farmer is going to use fence to make a pen shaped like cutting a circle in half and inserting a rectangle (like in the picture below). If the pen is to have an area of 100π, how should he make the pen to minimize the amount of fence needed? Find all dimensions of the pen and sketch a picture of it.

We begin by labeling the diagram as follows.



We can therefore calculate the total amount of fencing used as

$$T = 2x + 2\pi\left(\frac{y}{2}\right) = 2x + \pi y$$

We know that the area of the pen is given by $A = xy + \pi\left(\frac{y}{2}\right)^2$. Since the pen needs to have an area of 100π this means that $100\pi = xy + \pi\left(\frac{y}{2}\right)^2 \rightarrow \frac{100\pi - \frac{\pi}{4}y^2}{y} = x \rightarrow x = \frac{100\pi}{y} - \frac{\pi}{4}y$.

With this information we can state that $T = 2x + \pi y \rightarrow T = 2\left(\frac{100\pi}{y} - \frac{\pi}{4}y\right) + \pi y = 200\pi y^{-1} + \frac{\pi}{2}y$

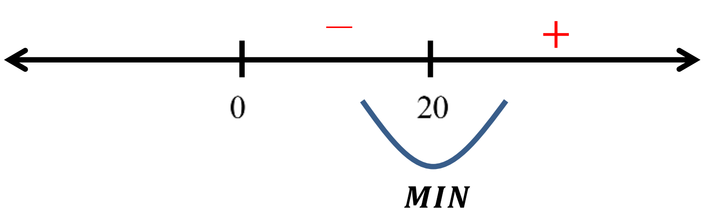
We then calculate $T' = \frac{-200\pi}{y^2} + \frac{\pi}{2} = \frac{\pi y^2 - 400\pi}{2y^2}$

We see that

$T' = 0$ when $\pi y^2 - 400\pi = 0 \rightarrow y = 20$

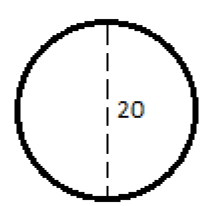
T' is undefined when $2y^2 = 0 \rightarrow y = 0$

We put these values on a number line to determine where T has a minimum.



Note that we aren't consider with values below 0 since these are not possible radius lengths.

So, we have that if $y = 20$ then the total amount of fencing used would minimized. This implies that $x = x = \frac{100\pi}{20} - \frac{\pi}{4}(20) = 5\pi - 5\pi = 0$ which means that the pen would ultimately be in the shape of a circle.



2. For each $x > 0$, a triangle is formed with vertices $(0,0)$, $(x,0)$ and $(x, f(x))$ where $f(x) = \dfrac{10}{x^2 + 3x + 9}$ is the function given below. What is the value of x which results in the triangle of largest area? What is this largest area?

We start by drawing a picture of the scenario.
We want to maximize the area of a triangle so we calculate that the area of a triangle here would be given by:

$$A = \frac{1}{2}xy = \frac{1}{2}x\left(\frac{10}{x^2+3x+9}\right) = \frac{5x}{x^2+3x+9}$$

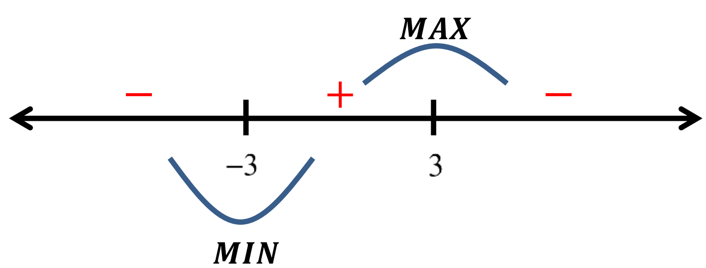
We then calculate

$$A' = \frac{(x^2+3x+9)(5) - 5x(2x+3)}{(x^2+3x+9)^2} = \frac{45-5x^2}{(x^2+3x+9)^2} = \frac{5(9-x^2)}{(x^2+3x+9)^2}$$

We see that

$A' = 0$ when $x = \pm 3$

A' is never undefined since $x^2 + 3x + 9 \neq 0$ for any value of x (Note $\sqrt{b^2 - 4ac} < 0$).

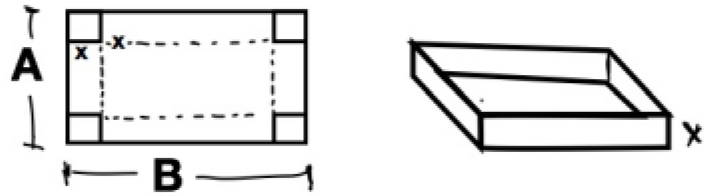


Remember that we are only looking at values of $x > 0$, so we can conclude that the absolute maximum of the area of the triangle would occur at $x = 3$.

The area of this triangle would be $A = \dfrac{5(3)}{(3)^2 + 3(3) + 9} = \dfrac{15}{27} = \dfrac{5}{9}$

3. A rectangular box, without a top, is to be made by cutting squares out of corners of a sheet of carboard and folding up the sides (like in the diagram below). Find the maximum volume of such a box from a sheet measuring 4 meters by 4 meters.

We see that the dimensions of the box are $(4-2x)$, $(4-2x)$, and x.
This implies that the volume of the box is given by

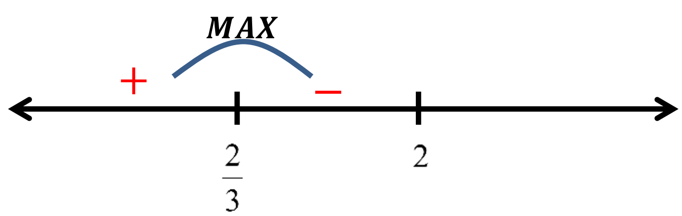


$$V=(4-2x)(4-2x)x=4x(2-x)^2=4x\left(4-4x+x^2\right)=4x^3-16x^2+16x$$

Therefore, $V'=12x^2-32x+16$

Clearly V' is never undefined.

$V'=0$ when $4\left(3x^2-8x+4\right)=0 \;\rightarrow\; 4(3x-2)(x-2)=0 \;\rightarrow\; x=\frac{2}{3}\;\&\; x=2$.

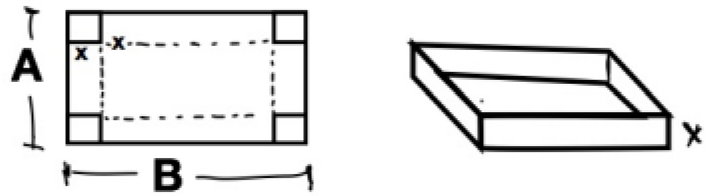


So, there is a maximum volume when $x=\frac{2}{3}$.

Therefore, the maximum value of the box is $V=\left(4-2\left(\frac{2}{3}\right)\right)^2\left(\frac{2}{3}\right)=\left(\frac{8}{3}\right)^2\frac{2}{3}=\frac{128}{27}$

4. A rectangular box, without a top, is to be made by cutting squares out of corners of a sheet of carboard and folding up the sides (like in the diagram below). Show that the maximum volume of such a box from a sheet measuring A meters by B meters occurs when the squares cut out have side lengths

$$x = \frac{(A+B) - \sqrt{(A+B)^2 - 3AB}}{6}$$



As shown in the diagram we will assume that $B > 0$

We see that the dimensions of the box are $(A-2x)$, $(B-2x)$, and x.

This implies that the volume of the box is given by

$$V = (A-2x)(B-2x)x = (AB - 2xA - 2xB + 4x^2)x = 4x^3 - 2x^2(A+B) + ABx$$

Therefore, $V' = 12x^2 - 4x(A+B) + AB$

Clearly V' is never undefined.

$V' = 0$ when $12x^2 - 4x(A+B) + AB = 0$

By the Quadratic Formula this occurs when

$$x = \frac{-(-4(A+B)) \pm \sqrt{(-4(A+B))^2 - 4(12)(AB)}}{2(12)}$$

$$= \frac{4(A+B) \pm \sqrt{16(A+B)^2 - 48AB}}{24}$$

$$= \frac{4(A+B) \pm 4\sqrt{(A+B)^2 - 3AB}}{24}$$

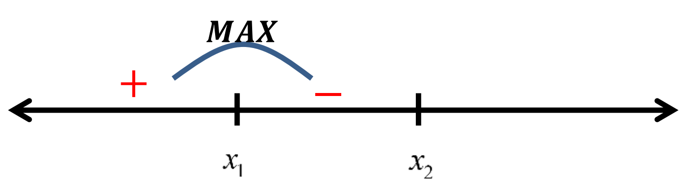
$$= \frac{(A+B) \pm \sqrt{(A+B)^2 - 3AB}}{6}$$

Notice there are two different values of x here:

$$x_1 = \frac{(A+B) - \sqrt{(A+B)^2 - 3AB}}{6} \quad \& \quad x_2 = \frac{(A+B) + \sqrt{(A+B)^2 - 3AB}}{6}$$

(Note these values only exist if $(A+B)^2 - 3AB > 0 \;\rightarrow\; (A+B)^2 > 3AB \;\rightarrow\; \frac{1}{3}(A+B)^2 > AB$)

We put these values on a number We put these values on a number line and check for where there are maxes and mins.



Note that $\frac{(A+B)}{6}$ is between x_1 and x_2. At this value we see that

$$\begin{aligned}
V' &= 12\left(\frac{A+B}{6}\right)^2 - 4\left(\frac{A+B}{6}\right)(A+B) + AB \\
&= \frac{1}{3}(A+B)^2 - \frac{2}{3}(A+B)^2 + AB \\
&= AB - \frac{1}{3}(A+B)^2
\end{aligned}$$

Based on our previous findings we can assuming that this value is negative.

Also note that $\frac{-\sqrt{(A+B)^2 - 3AB}}{6}$ is less than x_1. At this value we see that

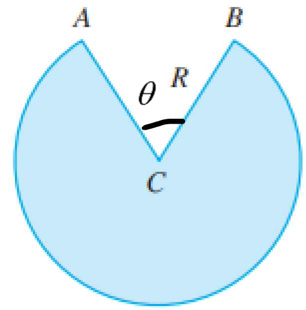
$$\begin{aligned}
V' &= 12\left(\frac{-\sqrt{(A+B)^2 - 3AB}}{6}\right)^2 - 4\left(\frac{-\sqrt{(A+B)^2 - 3AB}}{6}\right)(A+B) + AB \\
&= \frac{1}{3}\left[(A+B)^2 - 3AB\right] + \frac{2}{3}(A+B)\sqrt{(A+B)^2 - 3AB} + AB \\
&= \frac{1}{3}(A+B)^2 - AB + \frac{2}{3}(A+B)\sqrt{(A+B)^2 - 3AB} + AB \\
&= \frac{1}{3}(A+B)^2 + \frac{2}{3}(A+B)\sqrt{(A+B)^2 - 3AB}
\end{aligned}$$

Since $A > 0$ & $B > 0$ this value must also be positive.

Note that the maximum value that x can be is $\frac{B}{2}$. However, at this value the width of the box will equal 0 and so we can determine that even if the volume function increases after x_2 it can only increase to a value of 0 at best.

This shows that there must be a maximum value at $x_1 = \frac{(A+B) - \sqrt{(A+B)^2 - 3AB}}{6}$

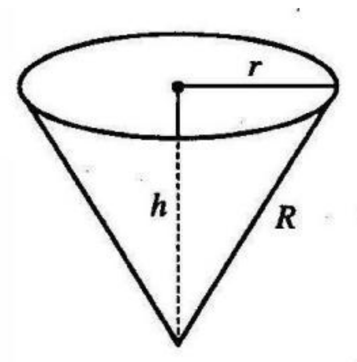
5. A cone-shaped drinking cup is made from a cicular piece of paper of radius R by cutting out a sector and joining the edges of CA and CB. Find the angle θ that is needed in order to maximize the capacity of such a cup.



We want to maximize the volume of the resulting cone.

Therefore we want to use the equation $V=\frac{1}{3}\pi r^2 h$.

Based on the picture of the resulting cone we can see that $h^2+r^2=R^2$ and therefore, $r^2=R^2-h^2$.

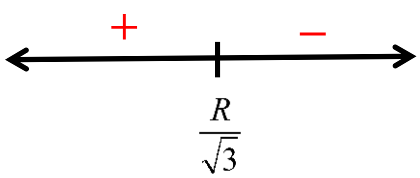


So we have that $V=\frac{1}{3}\pi\left(R^2-h^2\right)h=\frac{\pi}{3}\left(R^2h-h^3\right)$.

Note here that R is a constant and so the volume equation here is completely in terms of the variable h.

We calculate that $V'=\frac{\pi}{3}\left(R^2-3h^2\right)$.

$$V'=0 \text{ when } R^2-3h^2=0 \;\rightarrow\; h=\sqrt{\frac{R^2}{3}}=\frac{R}{\sqrt{3}}$$



(we tested the values of $h=0$ & $h=R$)

So we have a maximum volume when $h=\frac{R}{\sqrt{3}}$ and $r^2=R^2-\left(\frac{R}{\sqrt{3}}\right)^2 \;\rightarrow\; r=\sqrt{\frac{2}{3}}R$.

Now we turn to find the angle of θ.

We see from the first picture that the arc between A and B (over θ) is given by $arc\,length=\theta R$.

Also, the circumference of the existing part of the blue circle above must be equal to the circumference of the circular base of the resulting cone. From this we can say that

$$2\pi R-R\theta=2\pi r \;\rightarrow\; \theta=\frac{2\pi R-2\pi r}{R}=2\pi\left[1-\frac{r}{R}\right]$$

We then substitute the value of r that we have for when the maximum volume occurs and we find that

$$\theta=2\pi\left[1-\frac{r}{R}\right]=2\pi\left[1-\frac{\sqrt{\frac{2}{3}}R}{R}\right]=2\pi\left[1-\sqrt{\frac{2}{3}}\right]rad=360\left[1-\sqrt{\frac{2}{3}}\right]^{\circ}\approx 1.15rad\approx 66.06^{\circ}$$